

Define functional dependency!

List the Armstrong axioms and state the below statement. (6 points)

$\{WX \rightarrow Y, X \rightarrow Z, Z \rightarrow WY\} \models \{X \rightarrow Y\}$

A functional dependency $X \rightarrow Y$, between two sets of attributes specifies a constraint that any two tuples $T1$ and $T2$ that have $T1[x] = T2[x]$. They must also have $T1[y] = T2[y]$.

Armstrong's Axioms:

1. The Reflexive Rule: If Y is a proper subset of X , then $X \rightarrow Y$

2. The Augmentation Rule: $\{X \rightarrow Y\} \models XZ \rightarrow YZ$

3. The Transitivity Rule: $\{X \rightarrow Y, Y \rightarrow Z\} \models X \rightarrow Z$

Demonstration:

1. $X \rightarrow Z$

2. $Z \rightarrow WY$

3. $X \rightarrow WY$ (transitivity)

4. Y is a subset of W . $WY \rightarrow Y$ (reflexivity)

5. $\{X \rightarrow WY, WY \rightarrow Y\} \models X \rightarrow Y$ (transitivity)

Define functional dependency!

List only the Armstrong axioms and state the below statement using only them! (6 points)

$\{WX \rightarrow YZ, X \rightarrow Z, Y \rightarrow Q\} \models \{WX \rightarrow Z\}$

Demonstration:

1. $\{X \rightarrow Z\} \models WX \rightarrow WZ$ (augmentation)

2. $\{Z \subseteq WZ\} \models WZ \rightarrow Z$ (reflexivity)

3. $\{WX \rightarrow WZ, WZ \rightarrow Z\} \models WX \rightarrow Z$ (transitivity)

An embassy records details of interviews of visa applicants in the table below. Interviews are conducted by members of staff in some of the embassy rooms. In any given day, a member of staff tends to use the same room throughout that day. An applicant cannot have two interviews in the same day.

applicantNo	interviewDate	interviewTime	staffNo	roomNo
AP1	13-May	10:30	S5	R101
AP2	13-May	12:00	S5	R101
AP3	13-May	12:00	S9	R200
AP2	22-Sep	10:30	S5	R200

- Explain the term candidate key. (2 points)
- List several candidate keys for the above (explain why it is a key). (2 points)

A candidate key is the minimal superkey, an attribute or a set of attributes needed to uniquely identify a tuple.

1. (applicantNo, InterviewDate)

- **Uniqueness:** Each combination of applicantNo and InterviewDate is unique. This ensures that no two records have the same applicantNo and InterviewDate combination, thus uniquely identifying each row.
- **Minimality:** No subset of this combination (neither applicantNo alone nor InterviewDate alone) can uniquely identify a record. Therefore, this combination is minimal.

3. (InterviewDate, InterviewTime, roomNo)

- **Uniqueness:** Each combination of InterviewDate, InterviewTime, and roomNo is unique. This ensures that no two records have the same InterviewDate, InterviewTime, and roomNo combination, thus uniquely identifying each row.
- **Minimality:** No subset of this combination (e.g., InterviewDate and InterviewTime, InterviewDate and roomNo, or InterviewTime and roomNo) can uniquely identify a record. Therefore, this combination is minimal.

From an SQL user's perspective, does the relational model provide logical and physical data independence? Why? (2 points)

Yes, the relational model provides both logical and physical data independence from an SQL user's perspective. Logical data independence allows changes to the conceptual schema without affecting SQL queries and applications. Physical data independence enables changes to the physical storage without altering the logical schema. This separation simplifies database management and ensures stability for users and applications.

Define functional dependency!

List the Armstrong axioms and state the below statement. (6 points)

$\{X \rightarrow Y, Z \rightarrow W\} \models \{XZ \rightarrow WY\}$

Demonstration:

$\{X \rightarrow Y, Z \rightarrow W\} \models \{XZ \rightarrow WY\}$ (composition)

With reference to a sample relation of your own choosing, explain and discuss the following relational model terminology, including its function and any related concepts.

[A good diagram showing your sample relation is required for full mark.]

- **Tuple**
- **Attribute**
- **Domain**
- **Degree**
- **Cardinality**

Each item is equally weighted. (15 points)

Consider a relation named Employee with the following schema:

Employee ("EmpID", "Name", "Age", "Department")

Tuple means row. Each row is uniquely identified by a primary key.

For example: ("E001", "Alice", 30, "HR") is a tuple in the "Employee" relation.

Attribute means column. Each column must have a unique name within that table (relation) and can be isolated across tables by qualifying the column name with the table name.

For example: "EmpID", "Name", "Age" and "Department" are attributes of the "Employee" relation.

The domain specifies what are acceptable data values within a column (and by implication the acceptable operations on that column's data).

For example: The domain of "EmpID" could be the set of all valid employee IDs such as strings following the pattern "E####". Or the domain of "Age" could be the set of all valid ages, typically integers ranging from 18 to 65.

The degree of a table is the width or number of columns and is part of the union compatibility criteria.

For example: The degree of the "Employee" relation is 4 because it has four attributes.

The cardinality of a table is simply the depth or number of rows in the table.

For example: "Employee" relation has 3 tuples, so the cardinality is 3.

A company uses the table below to record details of staff. Each staff has up to three qualifications:

StaffID	StaffName	Qualifications
S01	Ibanga	BSc, MSc, PhD
S02	Kumar	BSc, MSc
S03	Grant	BSc, PhD

a. Explain why this table is not in "First Normal Form" (1NF). (1 point)

b. Show how this table can be transformed into 1NF tables. Give two possible solutions. (2 points)

a. The attribute "qualifications" is multi-valued on the PK, hence the table is not in 1NF.

b. Solution 1. Students (StaffID, StaffName, Qualification1, Qualification2, Qualification3)

Solution 2.

Students (StaffID, StaffName) (2 marks)

Qualifications (StaffID *, Qualification) (2 marks)

Alternatively, students can use a third table (QualID, Qualification)

Given the following relational schema:

Jedi-Teams (master, apprentice)

Jedi (name, side, home-planet)

Government (leader, planet, position)

Inhabitants (specie, planet)

Write a relational algebra expression which will query (by listing their names) to find all “wookie” (which is specie) Jedi on the “light” side of the force! (4 points)

[SQL expression for half mark]

So, the final relational algebra expression is:

$\pi_{\text{name}} (\sigma_{\text{side}='light'}(\text{Jedi}) \bowtie_{\text{Jedi.home-planet}=\text{Inhabitants.planet}} \sigma_{\text{specie}='wookie'}(\text{Inhabitants}))$

SQL Expression

The corresponding SQL query would be:

sql

Copy code

```
SELECT J.name
FROM Jedi J
JOIN Inhabitants I ON J.home-planet = I.planet
WHERE J.side = 'light' AND I.specie = 'wookie';
```

a) In the database context, what do we mean by redundant data? (1 point)

b) Why might it be a good idea to have redundant data in a database? (2 points)

c) Why might it be a bad idea to have redundant data in a database? (2 points)

a) Redundant data in a database refers to the repetition of the same data in multiple places. This duplication can occur within a single table or across multiple tables, leading to multiple copies of the same piece of information.

b) Data Availability: Redundancy can increase data availability and reliability. If one instance of the data becomes corrupted or unavailable, other copies can still be used, enhancing fault tolerance.

c) Data Inconsistency: Redundant data can lead to data inconsistency. If one copy of the data is updated while others are not, it can result in conflicting information within the database. This makes maintaining data integrity more challenging.

The table shown below stores details of students and the overall grade each student obtained in different modules. The Primary Key is (StudentID, ModuleID).

StudentID	StudentName	ModuleID	ModuleName	Grade
S001	Smith	M01	Java	A
S001	Smith	M02	Databases	B
S002	Ford	M01	Java	B

(i) Which Normal Form does the above table violate and why? (2 points)

(ii) Give an example of an update anomaly and an example of a delete anomaly that may occur if the table is left un-normalised. Explain the problems that are caused. (4 points)

(iii) Show how you would normalise the table. (3 points)

(i) The table violates 2nd Normal Form. because there are two partial dependencies: $\text{StudentID} \rightarrow \text{StudentName}$ and $\text{ModuleID} \rightarrow \text{ModuleName}$.

(ii) Update anomaly: if a module is to be renamed, it is possible to mistype one of its instances, thus introducing an inconsistency. Deletion anomaly: if students are deleted, modules will also be deleted, resulting in a loss of modules' details.

(iii)

Student (StudentID, StudentName)

Module (ModuleID, ModuleName)

Results (StudentID *, ModuleID *, Grade)

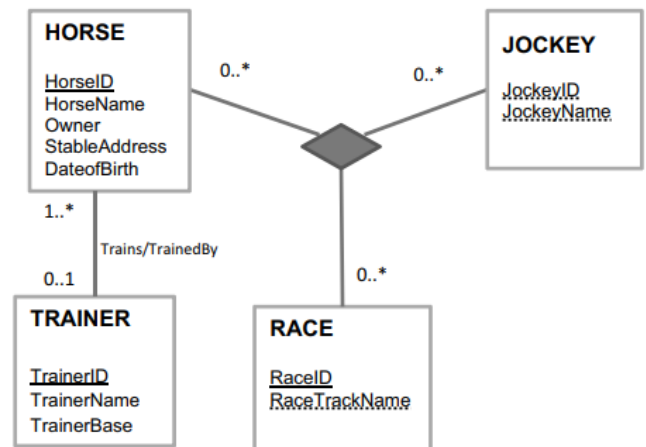
Refer to the ER model given in the figure. Assume that:

A trainer trains one or many horses and a horse is trained by no more than one trainer. A given horse competes in a particular race hidden by a particular jockey on a specific date and the position in which a horse finishes is recorded.

a) In the context of database design give a precise definition of an Entity Type(s). (2 points)

b) Explain using examples from the ER model (figure A1) the difference between a ternary relationship and a binary relationship. Why they are not the same? (4 points)

c) Explain how you would translate your modified ER model as a set of Tables, giving table names, column names and an indication of primary and foreign keys. (5 points)



a) Entity type describes an object, a noun, a class, any meaningful collection of objects of the same type having the same properties/attributes.

b) Ternary relationships have three participating entity types whereas binary have two. Example : results and trains respectively. An example will show diagrammatically examples of instances of horseID, jockeyID, RaceID all participating in the same relationship. Binary relationship instances have a set of instances that relate a horse to a particular trainer and a trainer related to many horses. Some instances of a horse may not have a (current) trainer.

c) Many to many relationships automatically introduce an associative entity that maps directly to a new table with a composite key consisting of fields that exist in the original tables forming the relationship. Hence the ternary relationship would be transformed into a table with a composite key comprising of the three identifiers from the three entity types forming the relationship. One to Many relationships map to two tables generally although technically this usually depends on the participation constraint on the one side.

Result Table

HorseID, JockeyID, ResultDate, RaceID, Position

Horse table

HorseID, horseName, Owner, DateOfBirth, StableAddress, TrainerID

Trainer table

TrainerID, TrainerName, TrainerBase...

Jockey table

jockeyID, jockeyName.

Given the following relational schema:

Driver (Name, LicenseNumber, Address)

CAR (LicensePlate, Type, Manufacturer)

OWNER (LicensePlate, LicenseNumber, Date_of_Purchase)

Write a relational algebra expression which gives all the drivers living in Debrecen and have a car! (4 points)

Write a relational algebra expression which gives all the drivers living in Debrecen and have a LEXUS car purchased in 2023! (4 points)

$\pi_{Name}((\sigma_{Address='Debrecen'}(Driver) \bowtie_{Driver.LicenseNumber=OWNER.LicenseNumber} OWNER) \bowtie_{OWNER.LicensePlate=CAR.LicensePlate} CAR)$

Using your own simple examples and suitable diagrams, explain and discuss its properties of the following Relational Algebra operations. For full marks, you must then supply an example SQL statement of how that concept is implemented.

a) **PROJECTION**

b) **THETA-JOIN**

c) **INTERSECT**

d) **SELECTION**

e) **UNION**

Each item is equally weighted.

- a. Projection is an operator that extracts specific columns from a relation.
- b. Theta-join combines tuples from two relations based on a condition.
- c. Intersect returns tuples that are present in both relations.
- d. Selection is an operator that extracts rows from a relation based on a condition.
- e. Union combines tuples from two relations, removing duplicates.

Consider the relation **REFRIG** (Model#, Year, Price, Manuf_planet, Color), which is abbreviated as **REFRIG** (M, Y, P, Mp, C), and the following set **F** of functional dependencies:

$F = \{ M \rightarrow Mp, \{ M, Y \} \rightarrow P, Mp \rightarrow C \}$

a) Evaluate each of the following as a candidate key for **REFRIG**, giving reasons why it can or cannot be a key: {M}, {M, Y}, {M, C}.

b) Based on the above key determination, state whether the relation **REFRIG** is in 3NF and in BCNF, giving proper reasons.

c) Consider the decomposition of **REFRIG** into:

$D = \{ R1 (M, Y, P); R2 (M, MP, C) \}$.

Is this decomposition lossless? Show why!

Consider the relation Employee_Department in the following Figure.

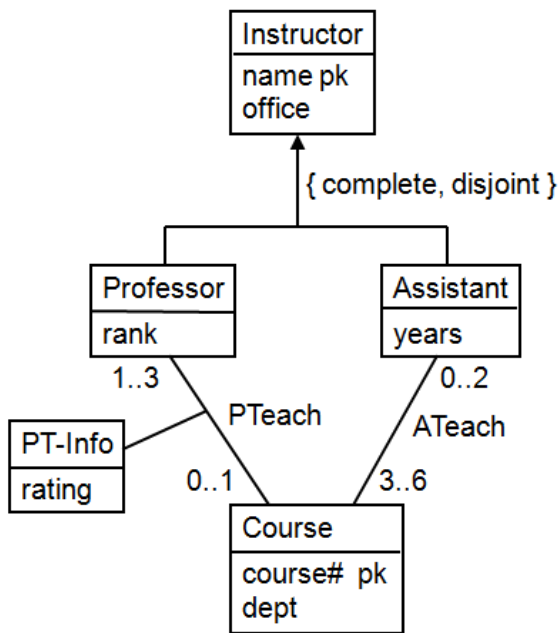
Ename	Enum	Bdate	Dnumber	Department	Dhomepage
John B	100	1965-01-03	2	Computing Science	www/it/cos
Smith L	105	1955-10-12	2	Computing Science	www/it/cos
James P	110	1962-03-03	1	Computer Systems	www/it/sys
Anna R	111	1976-11-23	4	Scientific Computing	www/it/sci
Pia S	112	1964-10-10	4	Scientific Computing	www/it/sci

The relation contains the following attributes:

Ename: Employee name, Enum: Employee number, Bdate: Employees birthdate, Dnumber: Department number, Dhomepage: Department homepage.

- What update anomalies may occur in Employee_Department? (2 points)
- List all the functional dependencies we have in this table! (2 points)
- In which normal form (NF) is the relation Employee_Department and why? (2 points)
- Show how you would normalise this table into 3NF. (4 points)
- Sketch an Entity - Relationship model for the data shown in Table. Show all the entities, relationships, and attributes!! (3 points)

Provide a set of relational tables for the high-level data model shown below. Identify primary and foreign keys in the tables. There are several possible automatic translations; use the translation for subclassing most appropriate for the specified properties. (8 points)



The following table shows a relation called **WAREHOUSE**.

Your assignment is as follows:

The warehouse database stores inventory information. A warehouse has one manager who has an employee number (unique) and a name. A warehouse is identified by its warehouse name (unique) and its address. Every part in the warehouse has a part number (unique), inventory date, quantity-on-hand, supplier name (one), and the delivery number, date, and quantity. One delivery could deliver several parts. The data is shown below:

manager-id	manager-name	warehouse-name	warehouse-address	part-no	inventory-date	qty-on-hand	supplier-name	delivery-no	delivery-date	delivery-qty
0001	John Duster	South	1001 St Paul St	A001	7/28/2015	5	Vollzo	D001	7/15/2016	5
0001	John Duster	South	1001 St Paul St	A002	7/23/2015	6	Vollzo	D001	7/15/2016	10
0001	John Duster	South	1001 St Paul St	A003	8/15/2016	2	Randste	D002	08-09-2016	20
0004	Marion Jones	North	1004 St Paul St	A002	7/26/2015	2	Vollzo	D009	7/20/2015	9
0004	Marion Jones	North	1004 St Paul St	A005	09-12-2015	8	Happag	D004	09-01-2015	15
0004	Marion Jones	North	1004 St Paul St	A006	04-01-2015	12	Panaso	D005	3/25/2015	20
0004	Marion Jones	North	1004 St Paul St	A007	03-12-2016	23	Whirlpo	D004	09-01-2015	30
0008	Hugh Dawson	East	1008 St Paul St	A008	08-08-2016	4	SMC	D006	08-01-2016	5
0008	Hugh Dawson	East	1008 St Paul St	A007	09-12-2016	20	Whirlpo	D006	09-06-2016	31

- Draw a relational schema and diagram for functional dependencies in the figure. (4 points)
- In what normal form is this relation and why? (2 point)
- Decompose **WAREHOUSE** into a set of 3NF relations. (4 points)
- What functional dependencies are existing in the figure. (4 points)
- Show the referential integrity constraints as well in these relations. (2 points)

Show and **USE** the advanced properties (inheritance, special type of attributes...) of the E-R model! Without using those, only half of the marks could be earned!

- Based on the exercise, using the same domain, extend it with (your own created) entities (People(Manager, Employee), Delivery, ...) and construct an E-R diagram for the **WAREHOUSE**.

The following table stores details of doctors, patients and dates of appointments. The Primary Key is (doctorID, patientID).

<u>doctorID</u>	doctorName	<u>patientID</u>	patientName	date
D01	Kumar	P02	Smith	10-August
D01	Kumar	P01	Ford	15-September
D02	Robinson	P02	Smith	10-August

- Explain why this table is not in 2NF. (2 points)
- Describe two types of anomalies that could be caused by update, insert or delete operations giving an example of each, with reference to the table above. (4 points)
- Transform table into 2NF. Show the structures of the resultant tables. (4 points)

Show and USE the advanced properties (inheritance, special type of attributes...) of the E-R model! Without using those, only half of the marks could be earned!

- Based on exercise 3, using the same domain, extend it with (your own created) entities (Staff, Applicant, ...) and construct an E-R diagram for the embassy. (10 points)
- Construct (transform into) appropriate relational tables for the above ER Diagram. (8 points)

Consider the following relation for published books:

BOOK (Book_title, Author_name, Book_type, List_price, Author_affil, Publisher)

Author_affil refers to the affiliation of author. Suppose the following dependencies exist:

Book_title $\rightarrow$ Publisher, Book_type

Book_type $\rightarrow$ List_price

Author_name $\rightarrow$ Author_affil

a) What normal form is the relation in? Explain your answer. (2 points)

b) Apply normalization until you cannot decompose the relations further. State the reasons behind each decomposition. (4 points)

The table below lists customer/car hire data. Each customer may hire cars from various outlets throughout Glasgow. A car is registered at a particular outlet and can be hired out to a customer on a given date.

carReg	make	model	custNo	custName	hireDate	outletNo	outletLoc
M565 OGD	Ford	Escort	C100	Smith, J	14/5/98	01	Bearsden
M565 OGD	Ford	Escort	C201	Hen, P	15/5/98	01	Bearsden
N734 TPR	Nissan	Sunny	C100	Smith, J	16/5/98	01	Bearsden
M134 BRP	Ford	Escort	C313	Blatt, O	14/5/98	02	Kelvinbridge
M134 BRP	Ford	Escort	C100	Smith, J	20/5/98	02	Kelvinbridge
M611 OPQ	Nissan	Sunny	C295	Pen, T	20/5/98	02	Kelvinbridge

- Identify the functional dependencies represented by the data shown in the table. State any assumptions you make about the data. (5 points)
- Using the functional dependencies identified in part(a), describe and illustrate the process of normalization by converting that table to 3NF relations. Identify the primary and foreign keys in your 3NF relations. (4 points)
- Sketch an E-R model for the data shown in the table. Show all the entities, relationships and attributes. (4 points)